Supplementary Step 7 Visualization Report

Overview

Metric	Value
Representative geometry session	PID001 / 2025-11-06_008
Included participant-sessions from Step 7	36
Included questions in Step 7 trial table	30
Abstract onset-locked epochs used	539
Concrete onset-locked epochs used	536
Epoch window	-3.5 s to 13.2 s around question onset
Baseline	-2.0 s to 0.0 s
Long-channel montage used in descriptive plots	22 pairs (44 chromophore channels)
Primary Step 7 duration-aware p-value	0.102262

Adaptation note. This report mirrors the Step 6 visualization bundle but anchors it to the Step 7 duration-aware cohort and primary model results. Because the Step 7 inferential model summarizes continuous fNIRS over variable-length windows, the descriptive topographic plots here remain onset-locked so that all trials share a common time base for direct visual comparison.

Rendering note. As in the previous supplementary visualization reports, the sensor-geometry figure uses the participant digitization point cloud plus source, detector, channel, and pair geometry rather than an fsaverage brain-surface rendering, because the local `pyvista/fsaverage` stack is not available in this environment.

Step 7 Main Figures Reused

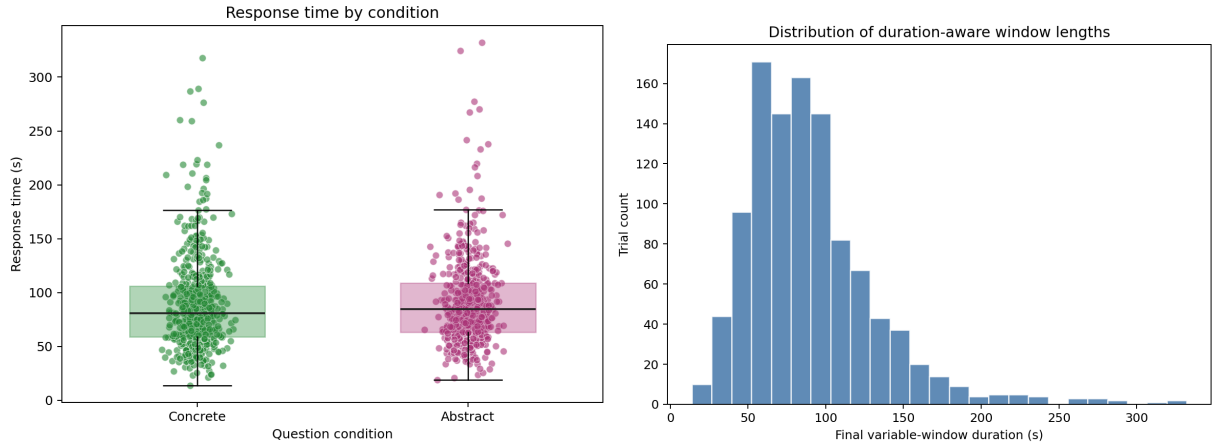

Figure 1: Step 7 duration diagnostics for response time by condition and the distribution of final variable-window durations.

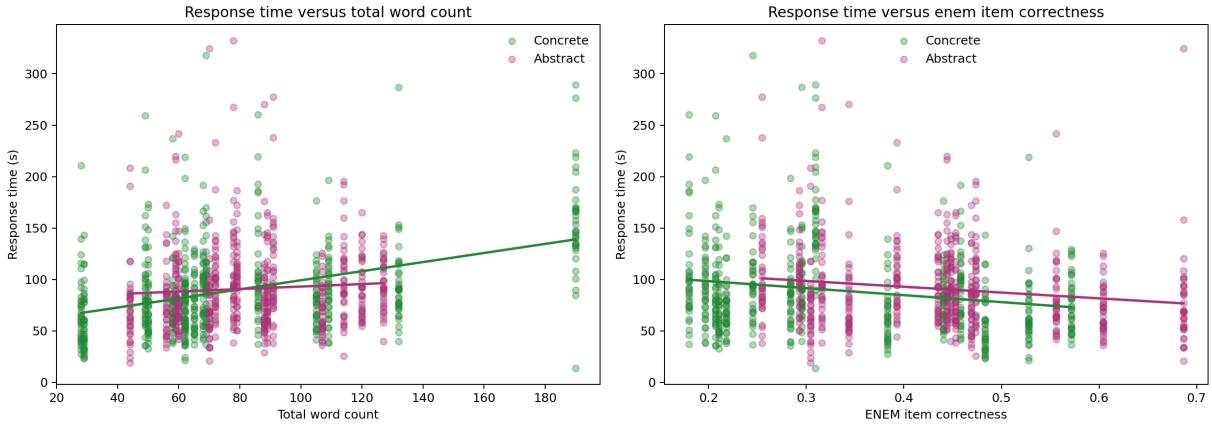

Figure 2: Step 7 response-time associations with item length and ENEM item correctness.

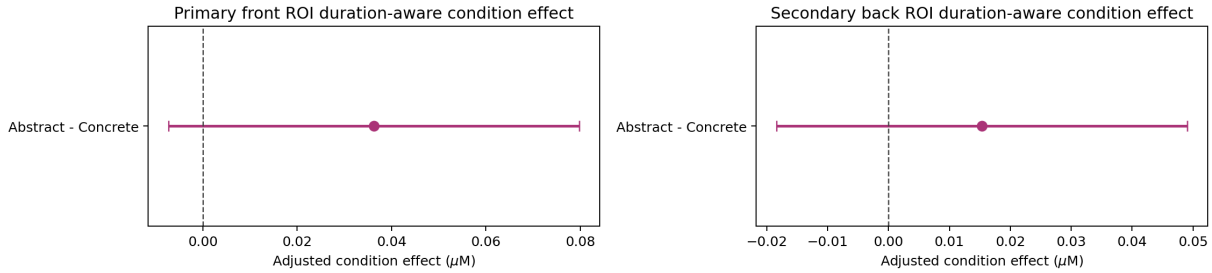

Figure 3: Step 7 duration-aware condition-effect summaries for the front and back ROI models.

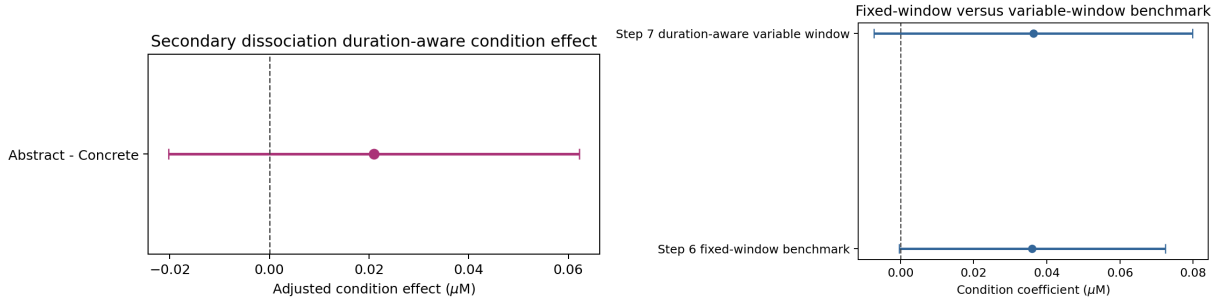

Figure 4: Step 7 dissociation effect and the direct benchmark against the Step 6 fixed-window model.

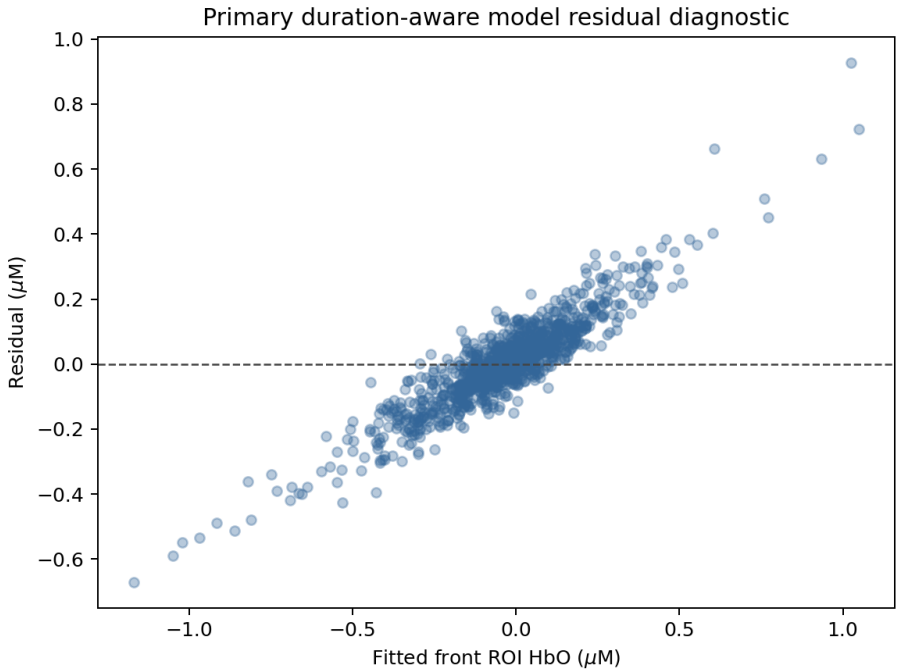

Figure 5: Residual diagnostic from the Step 7 primary duration-aware model.

Step 6-Style Supplementary Plots for Step 7

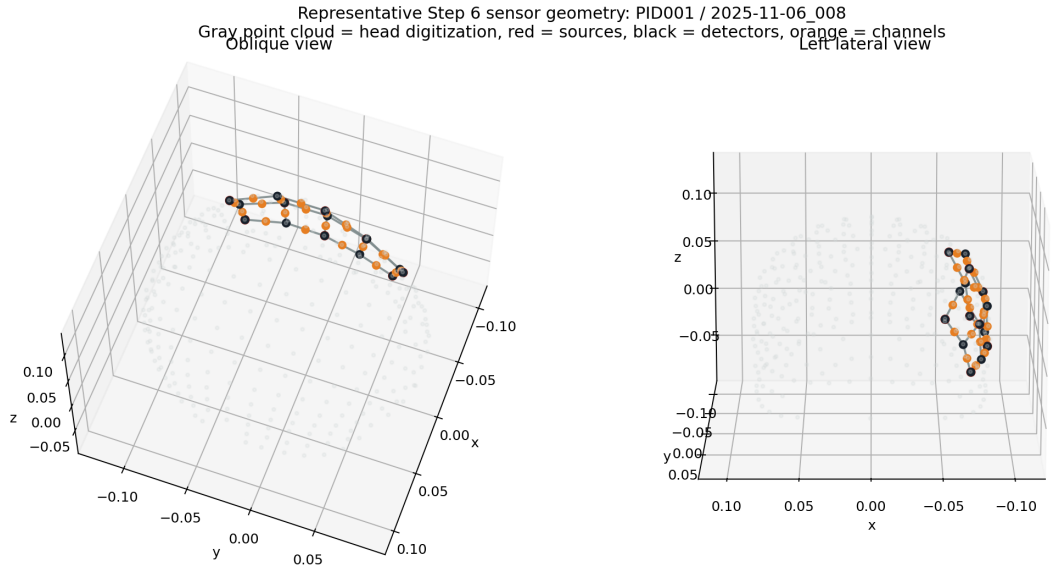

Figure 6: Representative optode, channel, and source-detector-pair geometry for one Step 7 included participant-session. Short pairs are shown with dashed orange lines.

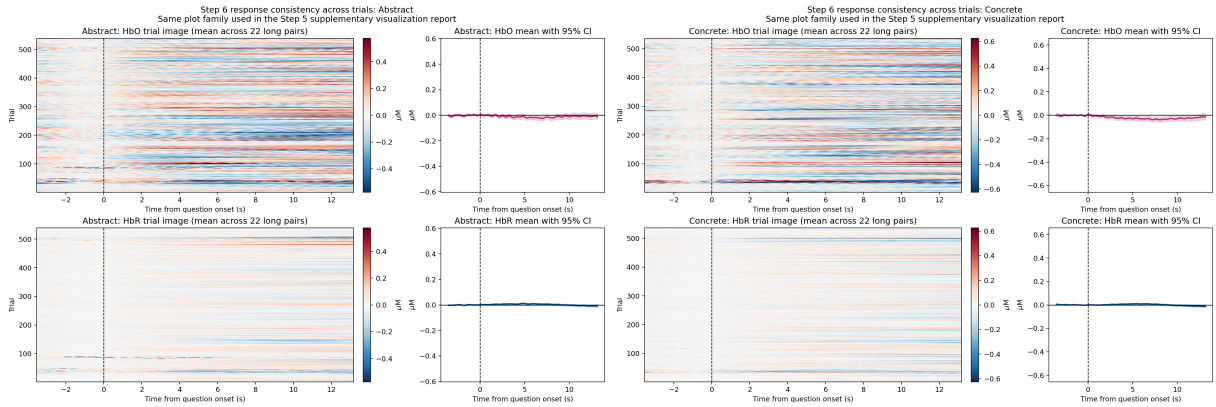

Figure 7: Condition-adapted trial-consistency image plots for the Step 7 included trial set.

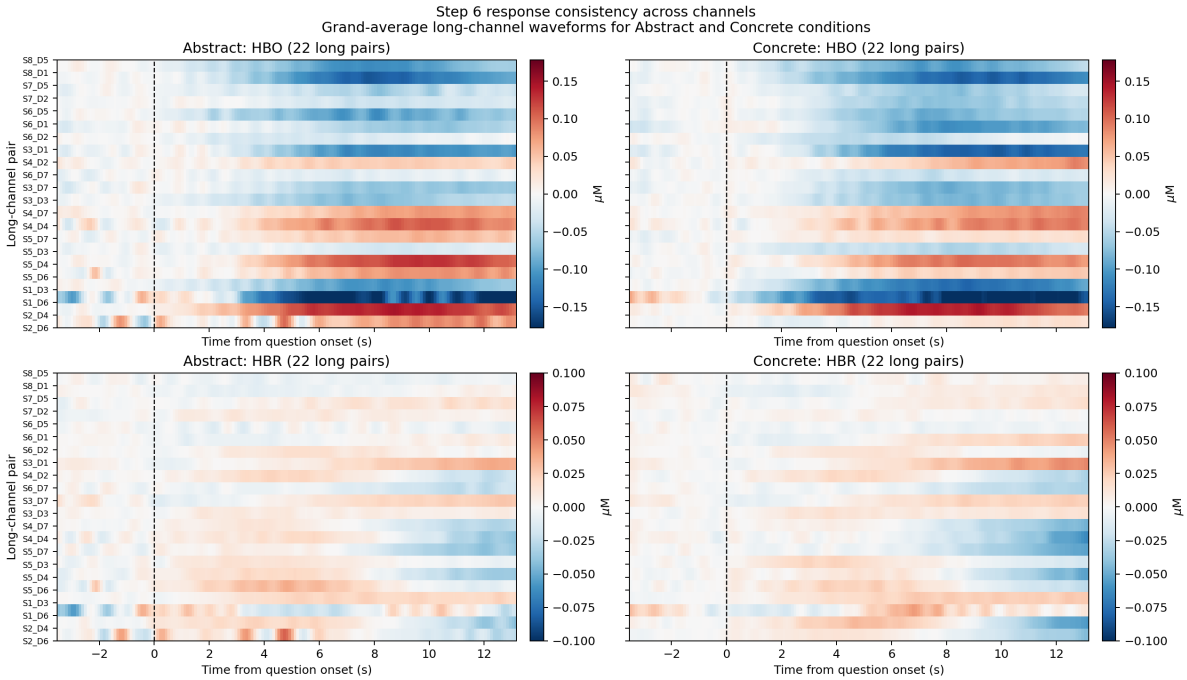

Figure 8: Grand-average long-channel response images across channels for the Abstract and Concrete conditions.

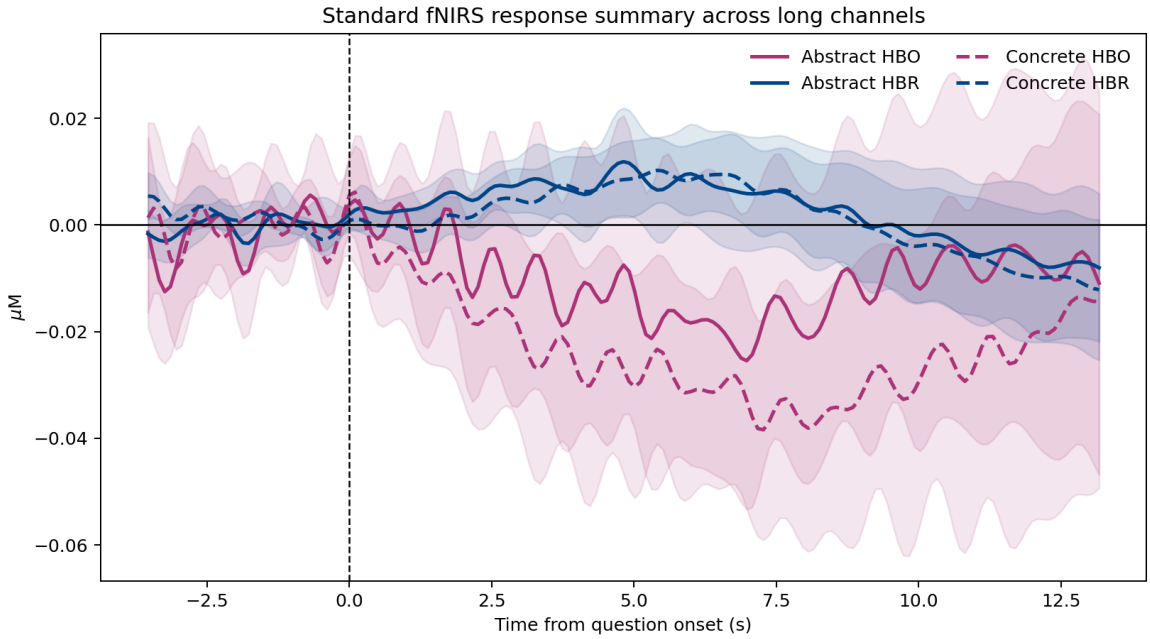

Figure 9: Standard fNIRS waveform comparison across long channels for the Step 7 included trial set.

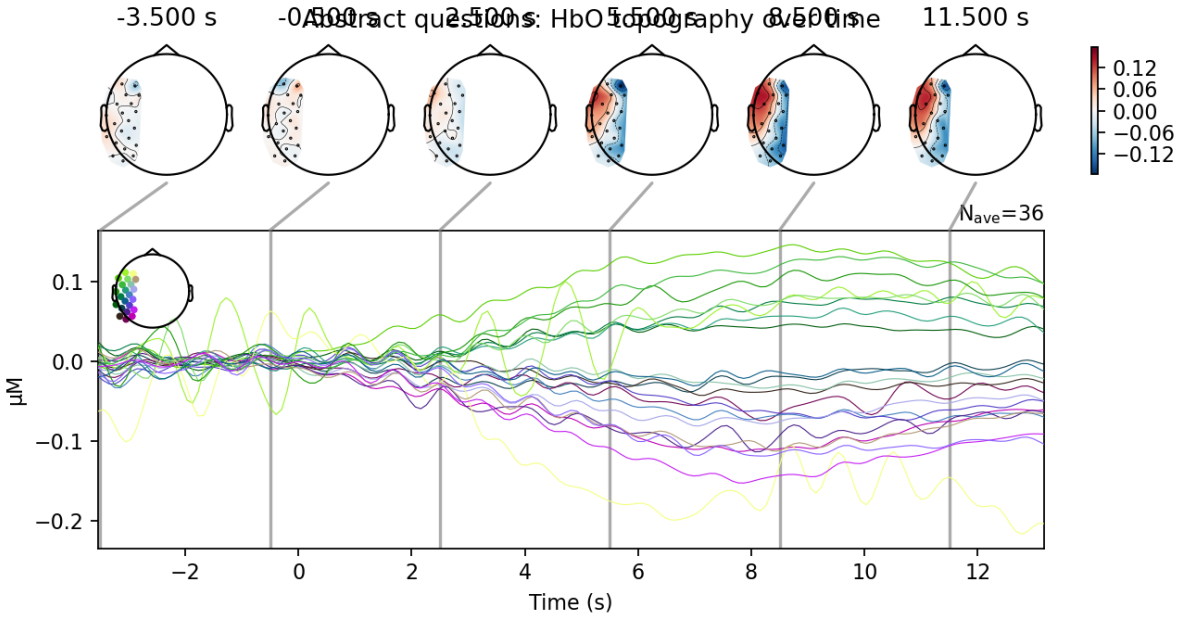

Figure 10: Grand-average HbO topography over time for Abstract questions using the Step 7 included trial set.

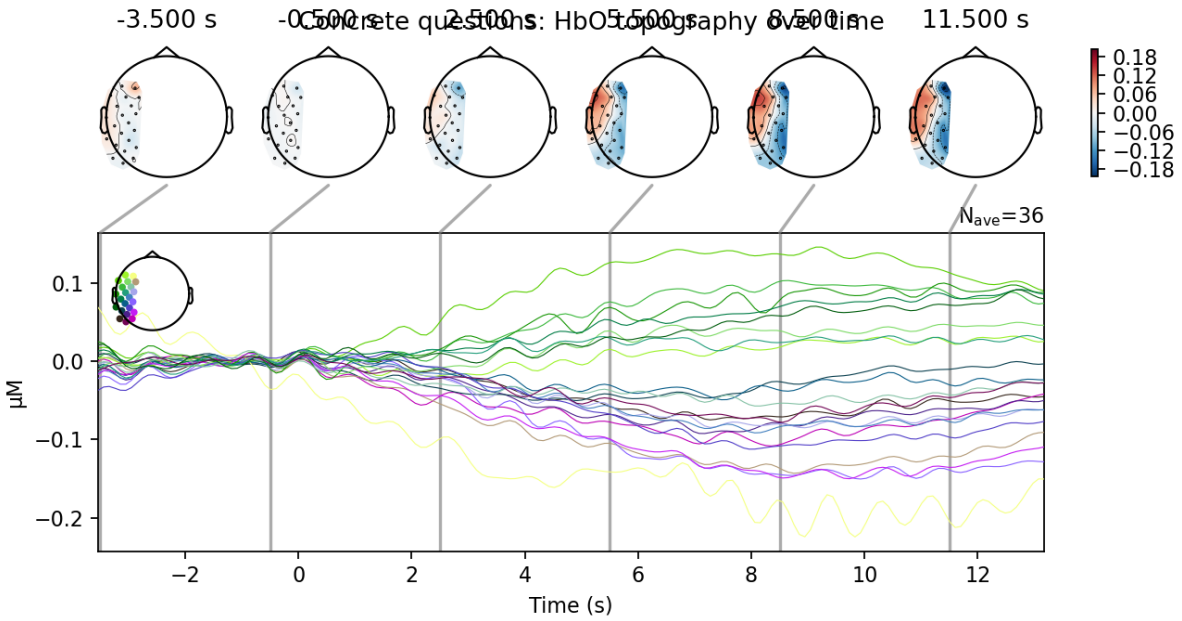

Figure 11: Grand-average HbO topography over time for Concrete questions using the Step 7 included trial set.

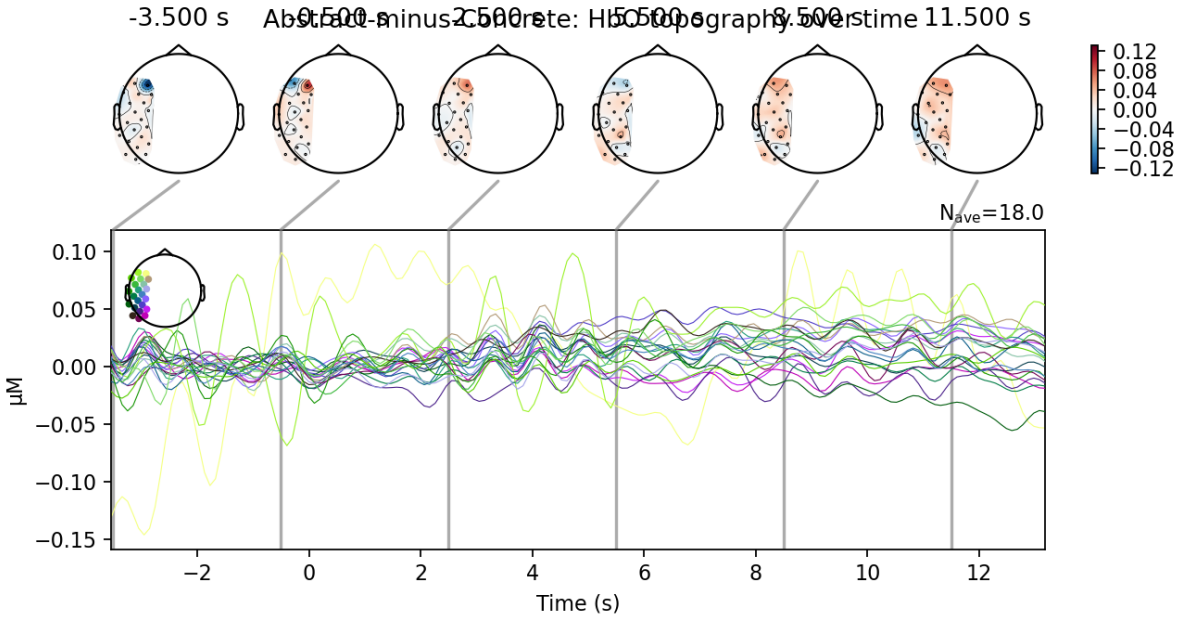

Figure 12: Grand-average Abstract-minus-Concrete HbO topography over time.

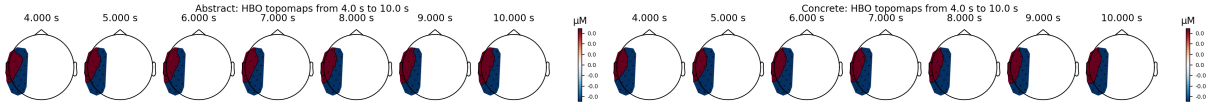

Figure 13: HbO topographic time series from 4.0 s to 10.0 s for the Abstract and Concrete conditions.

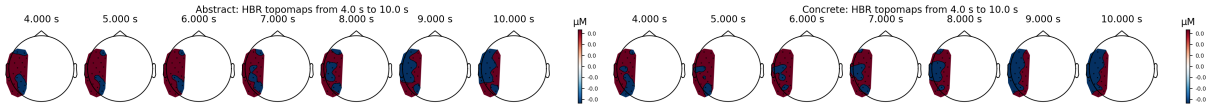

Figure 14: HbR topographic time series from 4.0 s to 10.0 s for the Abstract and Concrete conditions.

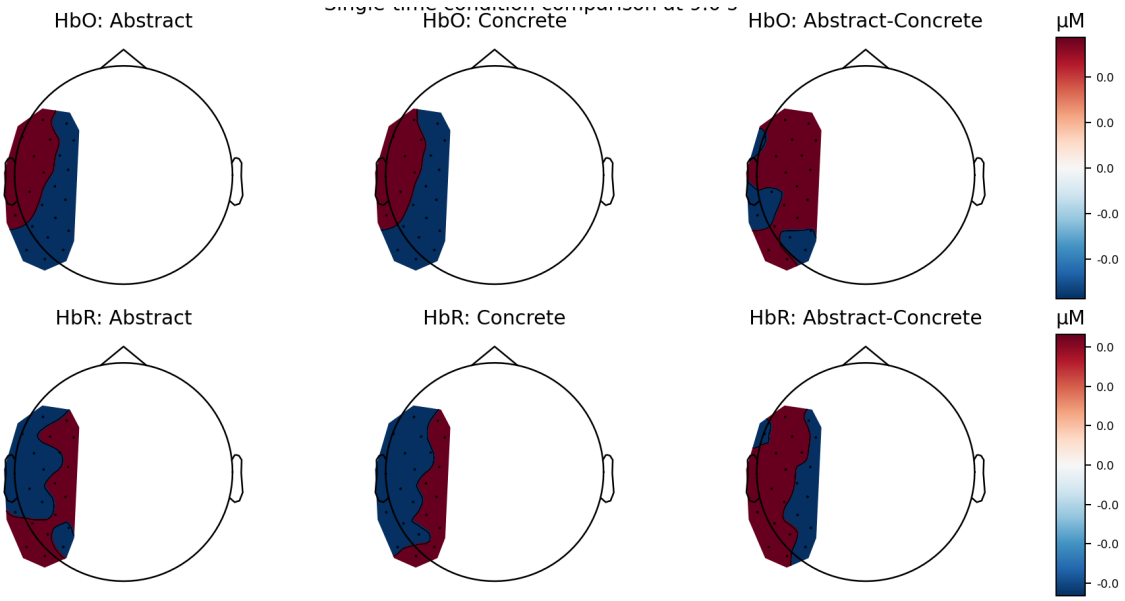

Figure 15: Single-time comparison at 9.0 s for Abstract, Concrete, and Abstract-minus-Concrete topographies for both HbO and HbR.

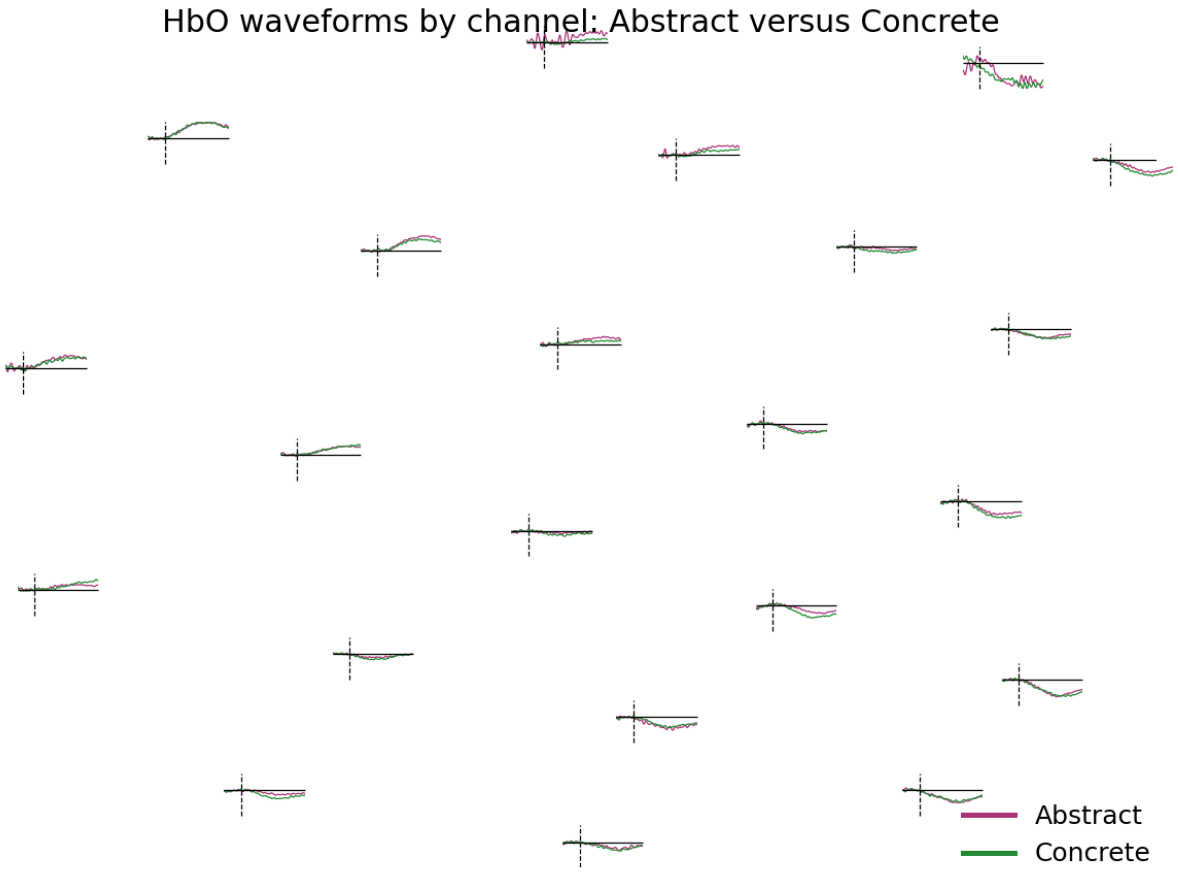

Figure 16: HbO waveforms by long channel, overlaid for Abstract and Concrete, to show what drives the descriptive topographic differences.